A Data-Driven Approach to Session-Based Recommendation using Graph-based Modeling.

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***Abstract*—Online retail platforms increasingly face situations where persistent user identity is incomplete or unavailable, making traditional user-profile-based recommendation less effective. This study investigates whether anonymous session-level behavior can be used to infer short-term user intent and support product recommendation in e-commerce. Using the Diginetica dataset, we formulate the task as a session-based next-item recommendation problem, where the system predicts the next product from a sequence of in-session item views. We compare three approaches: a popularity-based baseline, a co-occurrence baseline, and SR-GNN, a graph neural network model designed to capture sequential dependencies within sessions. The results show that SR-GNN substantially outperforms the baselines in HR@20, MRR@20, and catalog coverage, indicating its stronger ability to recommend relevant products from short-term behavioral signals. Additional simulation analysis suggests that improved recommendation accuracy may contribute to higher engagement, conversion potential, and average order value proxy. Overall, the study demonstrates how session-based recommendation can help e-commerce platforms better utilize anonymous traffic without relying on persistent user identity.**

***Index Terms*—Session-based Recommendation, Machine Learning, Marketing Analytics, Graph Neural Networks (GNN), User Intent Prediction**

I. Introduction

In the current e-commerce landscape, online retail platforms are increasingly operating in environments where persistent user identity is incomplete, unavailable, or difficult to collect. The shift toward a cookieless era further reduces the ability to rely on long-term user profiles for personalization. At the same time, asking customers to register, log in, or provide detailed personal information before browsing may introduce friction and negatively affect the customer journey. Therefore, a large portion of behavioral signals is generated within anonymous or short-lived sessions, where the user's immediate intent must be inferred directly from their recent interactions.

This context directly affects the design of recommender systems. Traditional recommendation approaches often rely on long-term user profiles, historical user-item interaction matrices, or rich contextual information to personalize product suggestions. However, when users remain anonymous, these methods become less effective because the system cannot fully observe who the customer is or what they have done in the past. Therefore, the key challenge is how to build a recommendation system that can still provide relevant product suggestions when user-level information is limited.

Nevertheless, anonymous traffic should not be treated as unusable data. Even when user identity is missing, the sequence of actions within a shopping session can still reveal meaningful short-term intent. The products that a customer views, clicks, or compares during a visit provide important signals about their current interests. This suggests that the recommendation problem should shift from relying mainly on long-term user identity to exploiting the behavioral patterns observed within each session.

In this study, we examine this challenge using the Diginetica E-commerce dataset. The dataset provides a suitable setting for studying recommendation in anonymous and session-based environments, where user profiles are limited but product interaction sequences are available. Our focus is to investigate how session-level behavior can be transformed into useful signals for recommending relevant products during the customer journey.

By positioning recommender systems at the center of the analysis, this study aims to show how data-driven marketing can move beyond descriptive customer analysis and support actionable product recommendation. The broader objective is to help E-commerce platforms better utilize anonymous sessions, improve the relevance of recommended products, and increase opportunities for user engagement and conversion.

II. Data Description and Preprocessing

A. Data description

This study uses the Diginetica dataset, which was originally released as part of the CIKM Cup 2016 challenge on e-commerce search personalization. The dataset captures real user browsing and purchasing behavior on an e-commerce platform over a period of approximately six months, from January 1, 2016 to June 17, 2016. It records how users interact with products — what they view, what they click on, and what they ultimately buy — making it a well-suited benchmark for session-based recommendation tasks.

The dataset consists of six files, each capturing a different aspect of the platform's data: item views, purchases, search queries, search clicks, product metadata, and product category mappings.

It is worth nothing that search queries and search clicks are treated as optional files in this project, as the recommendation task is built directly on users' in-session item viewing behavior, and search context is not required as part of the input.

B. Data preprocessing

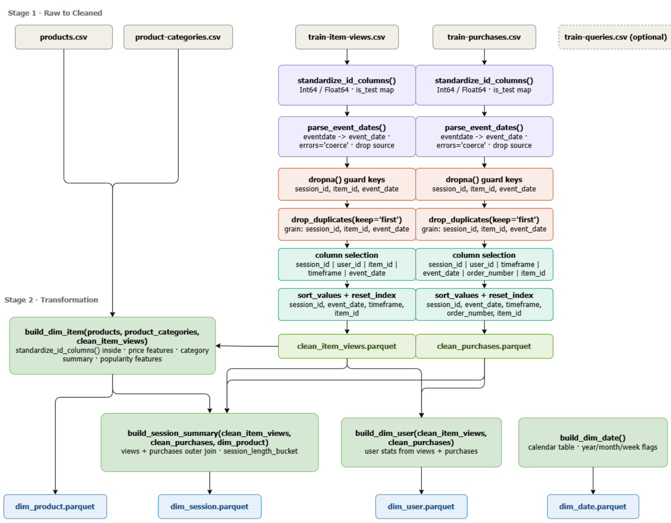

Fig. 1: Data preprocessing pipeline

The cleaning pipeline takes raw Diginetica data and prepares it through distinct steps.

The first thing the pipeline does is bring everything into a consistent shape. Column names are rewritten from camelCase and dot notation into snake_case, so the codebase speaks one language throughout. Identifier columns are cast to nullable integers (`Int64`) — a necessary choice because a large portion of users are anonymous and carry no `user_id`, and a regular integer type has no way to represent that absence. The raw event date string is parsed into a proper datetime, and `pricelog2` is cast to a nullable float.

With the basics in order, the pipeline builds a clean product reference table by joining product and category information into a single, one-row-per-product view. Because prices are stored on a log base-2 scale, a `price_proxy` column is added using the formula $2^{\texttt{pricelog2}} - 1$, giving a number closer to what the original price actually looked like. Category information is rolled up per item into a primary category, a category count, and a full list of category IDs. Each item also gets a view count pulled from the training data and is placed into one of six popularity buckets — `unviewed`, `1`, `2-5`, `6-20`, `21-100`, and `101+` — making it easy to reason about how well-known a product is.

The behavioral tables — item views and purchases — are then cleaned using the exact same steps. Any row missing a `session_id`, `item_id`, or `event_date` is dropped, since there is no meaningful way to use it. Duplicate rows across those three fields are removed, keeping only the first occurrence. What remains is sorted chronologically within each session, so that the natural order of a user's actions is preserved.

The final step pulls everything together into a session-level summary table, with one row per session. View events and purchase events are aggregated separately, capturing things like view count, number of unique items seen, purchase count, and estimated spend. The two sides are then merged with an outer join, so sessions that only contain views and sessions that only contain purchases are both kept. A `has_purchase` flag marks whether any transaction took place, and each session is assigned a length bucket — `0`, `1`, `2`, `3-5`, `6-10`, `11-20`, or `21+` — for easier filtering and analysis later on.

III. Exploratory Data Analysis (EDA)

This section presents empirical evidence from the dataset to confirm the urgency of building session-based recommendation systems.

A. Anonymous Sessions as a Major Traffic Segment

As show in **Figure 2**, traffic analysis highlights a critical challenge: the vast majority of users interact with the system without logging in. Specifically, **anonymous sessions** account for an overwhelming majority of **216K sessions (69.57%)**, which is more than double the volume of logged-in sessions. This prevalence of unidentified users creates a significant "blind spot" for traditional algorithms, such as **Collaborative Filtering**. Since these methods rely heavily on User IDs and long-term historical data, they are unable to effectively personalize experiences for over two-thirds of the traffic, resulting in missed opportunities to optimize revenue from the platform's most significant user base.

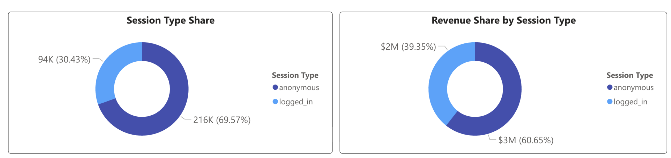

Fig. 2: Anonymous and identified session analysis

B. Revenue and Engagement Contribution of Anonymous Sessions

Contrary to the technical limitations in tracking, anonymous sessions serve as the primary economic engine of the platform, accounting for **60.65% of the total turnover ($3M)**. This empirical evidence refutes the common misconception that unidentified traffic primarily consists of "spam" or low-value interactions. While these users lack formal profiles, their intensive interaction patterns present a vital opportunity for engagement. Neglecting this segment would not only result in a substantial loss of potential revenue but also overlook the immense predictive intelligence embedded within the existing click-stream data.

C. Session Interaction and Conversion Potential

Figure 3 demonstrates a compelling correlation between session depth (number of interactions) and conversion efficiency. While the vast majority of sessions are short-lived—with over **250,000 sessions** containing only 1-5 clicks—these brief interactions yield the lowest conversion rate at approximately **3.9%**. In stark contrast, as users engage more deeply, the conversion probability climbs significantly, peaking at nearly **5.6%** for sessions with 11-20 interactions.

This trend underscores several key insights for session-based modeling:

- **Signal Strength:** Each subsequent click in a session serves as a high-intent signal, progressively refining the user's immediate interest.
- **Conversion Efficiency:** Sessions with extended interaction sequences result in substantially higher conversion rates compared to "shallow" sessions, despite their lower total volume.
- **Data Richness:** While purchase events are often scarce and sparse, the abundance of click-stream data in these deeper sessions provides a rich feature set for building accurate predictive models of subsequent behavior.

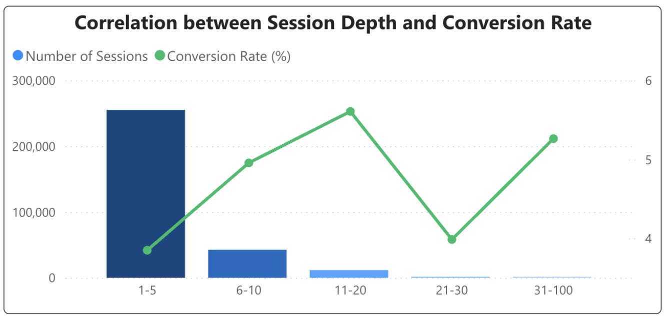

Fig. 3: Correlation between session depth and conversion rate

D. Key Findings from EDA

From the above analysis, we derive the core insight: "We may not know who they are (User Identity), but we know exactly what they are doing (User Intent) through interactions within the session." Deeply exploring real-time behavior patterns is key to personalizing the experience without relying on past user profiles.

IV. From Marketing Insights to Recommendation Problem

This section acts as a bridge, transforming business observations into a formal engineering task.

A. Task Motivation

From a marketing perspective, the high proportion of anonymous sessions identified in the EDA poses a significant challenge to customer retention and conversion rate optimization. Traditional entity-based approaches (such as Collaborative Filtering) fail to leverage this segment due to the absence of long-term User Profiles. Consequently, the focus must shift from identifying *"Who is buying?"* to predicting *"What is the next action after this sequence of behaviors?"*

By focusing on the session level, we can capture the user's immediate intent. View data is selected as the primary input signal because it is significantly richer and denser than purchase data, providing enough information to model user interest in real-time even without historical data.

B. Problem Formulation

We formulate the task as a **session-based next-item recommendation** problem. Let $\mathcal{I} = \{i_1, i_2, \ldots, i_m\}$ be the set of all unique items. Each session is represented as an ordered sequence of clicked products:

$$S_t = [i_1, i_2, \ldots, i_t]$$

where $i_j \in \mathcal{I}$ denotes the product clicked at step j. Given the current sequence S_t, the goal of the recommender system is to predict the actual next product $y_t = i_{t+1}$ by outputting a ranked list of the top 20 candidates:

$$\hat{Y}_t^{20} = [\hat{i}_1, \hat{i}_2, \ldots, \hat{i}_{20}]$$

To train the model, a single session $[i_1, i_2, i_3, i_4]$ is decomposed into multiple training instances to maximize data utility:

- $[i_1] \rightarrow i_2$
- $[i_1, i_2] \rightarrow i_3$
- $[i_1, i_2, i_3] \rightarrow i_4$

C. Success Measurement

The primary evaluation metric is **Hit Rate at 20 (HR@20)**, defined as:

$$HR@20 = \frac{1}{N} \sum_{n=1}^{N} 1(y_n \in \hat{Y}_n^{20})$$

where N is the total number of test instances.

Strategic Link: A high HR@20 value directly addresses the marketing objective of "customer stickiness." If the product the user intended to click is present in the top-20 recommendations, the system successfully maintains engagement, increases Click-Through Rate (CTR), and ultimately creates more opportunities for downstream conversion. Through this formulation, the marketing challenge of optimizing anonymous traffic is effectively transformed into a robust session-based recommendation problem.

V. Session-Based Recommendation Performance

A. Offline Accuracy Evaluation

Model Overview: The study compares three recommendation systems with increasing levels of complexity: the Popularity baseline (recommending the globally most popular items without session dependency), the Co-occurrence baseline (leveraging last-item transition statistics), and SR-GNN — a graph neural network model representing the entire view sequence as a directed graph to capture long-range dependency within sessions.

1) Core Evaluation Metrics: Three metrics were computed on 45,910 test examples using a temporal split (last 7 days).

TABLE I: Offline evaluation results on the test set

Model	HR@20	MRR@20	Coverage@20
Popularity	0.0101	0.0023	0.09%
Co-occurrence	0.3008	0.1081	96.7%
SR-GNN	0.5416	0.1832	93.6%

SR-GNN achieved HR@20 = 0.5416, indicating approximately one successful recommendation for every two recommendation attempts. Compared with the co-occurrence baseline, SR-GNN improved HR@20 by +80.0% and MRR@20 by +69.5%. Catalog coverage reached 93.6%, reflecting the model's ability to capture long-tail items rather than concentrating recommendations on a narrow product subset.

2) Performance Across Session Length Buckets:

TABLE II: HR@20 across session length buckets for Popularity, Co-occurrence, and SR-GNN

Session Length	Popularity	Co-occurrence	SR-GNN
1–3	0.014	0.324	0.580
4–5	0.013	0.322	0.560
6–9	0.010	0.301	0.546
10–20	0.008	0.285	0.525
20+	0.008	0.274	0.470

Table II shows that SR-GNN consistently outperformed both the co-occurrence and popularity baselines across all session length groups, with HR@20 ranging from 0.470 to 0.580, compared with 0.274–0.324 for the co-occurrence model. This result reflects the structural advantage of SR-GNN in modeling sequential session behavior. While popularity-based recommendation primarily prioritizes high-frequency products, the co-occurrence baseline relies only on last-item transitions and gradually degrades as session length increases. In contrast, SR-GNN graph propagation captures the entire interaction sequence, enabling the model to leverage accumulated purchase-intent signals across multiple interaction steps. For long sessions (10+ views), SR-GNN maintained stable performance, suggesting stronger capability in extracting behavioral signals from sessions associated with higher consideration and purchase intent.

B. Simulation Results

a) Counterfactual Simulation Framework:

Model Selection for Simulation: The impact analysis exclusively uses SR-GNN — the highest-performing model in the offline evaluation — as the representative state of "recommendation intervention." The remaining baselines are included only for comparison in Section 1 and are not used in the simulation framework.

The simulation applies a counterfactual bounce model. For each test session, the first view position where SR-GNN fails to place the target product within the top-20 recommendations is defined as the *bounce position*. All interactions occurring after this position — including purchase events — are removed. This process produces two datasets:

- **Before (Original):** observed real-world sessions without recommendation intervention.
- **After (With Model):** sessions truncated at the bounce position, representing the state where SR-GNN continuously guides user interaction.

Note. This framework represents an upper-bound counterfactual scenario rather than a realistic deployment forecast. All reported values should therefore be interpreted as directional proxies rather than causal evidence.

b) Conversion Rate by Session Length:

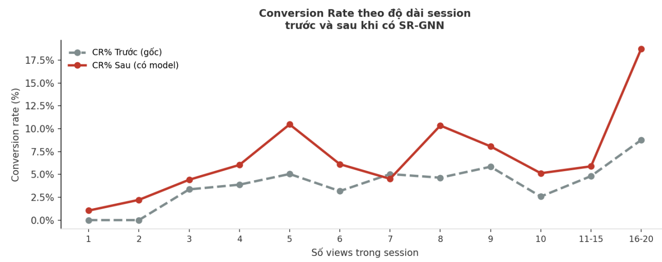

Fig. 4: Conversion rate by session length before and after SR-GNN intervention. X-axis: number of views per session; Y-axis: CR (%). Dashed line: baseline (before intervention).

Figure 4 presents conversion rate (CR%) across session lengths before and after SR-GNN deployment. The results show that SR-GNN improves CR% across most session-length groups, with the clearest gains observed in medium and long sessions. Specifically, for sessions with 5 views, CR% increased from 5.1% to 10.5%; for sessions with 8 views, CR% increased from 4.8% to 10.4%; and for sessions with 16–20 views, CR% increased from 9.1% to 18.4%. For short sessions (1–2 views), improvement remained limited, which is consistent with theoretical expectations: sessions that are too short provide insufficient behavioral signals for graph-based models to construct meaningful session representations. Overall, this trend suggests that SR-GNN becomes more effective when users generate sufficiently rich interaction histories within a session, supporting the argument that improved recommendation quality can translate into higher real-world purchase conversion.

c) Step-wise Conversion Funnel:

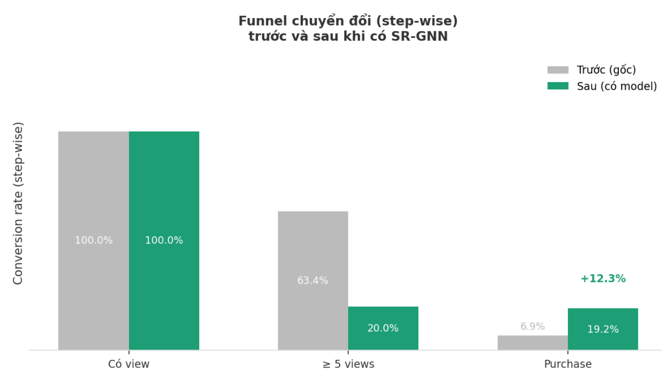

Fig. 5: Step-wise conversion funnel before and after SR-GNN intervention.

Figure 5 presents a three-stage funnel: (i) sessions with views (baseline 100%), (ii) sessions reaching $\geq$5 views, and (iii) purchases after reaching $\geq$5 views. The proportion of sessions reaching the $\geq$5-view stage decreased from 63.4% (Before) to 20.0% (After), reflecting the simulation mechanism:

sessions where SR-GNN generates early hits are shortened before crossing the 5-view threshold. More importantly, the conditional purchase rate after reaching $\geq$5 views increased from 6.9% to 19.2% (+12.3 percentage points). This result suggests that SR-GNN not only shortens sessions for easy-match cases but also improves recommendation quality for engaged, high-intent users.

d) Session Segmentation: Frequency $\times$ Purchase:

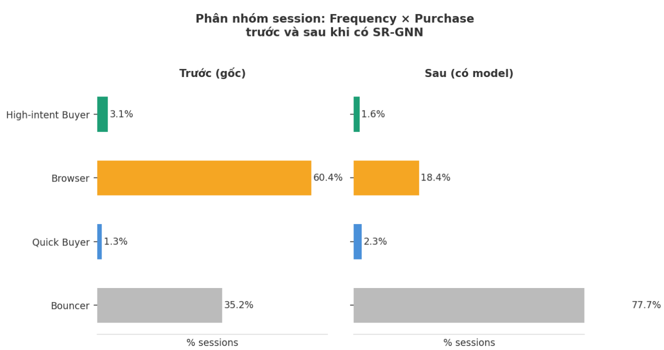

Fig. 6: Session segmentation by interaction frequency and purchase outcome, before (left) and after (right) SR-GNN intervention. Frequency threshold: 5 views.

Figure 6 applies a two-dimensional segmentation framework — analogous to a simplified RFM structure — based on interaction frequency (5-view threshold) and purchase outcome. The “Before” panel reveals the typical structure of a high-friction platform: 60.4% of sessions belong to the Browser segment ($\geq$5 views without purchase), while only 3.1% reach the High-intent Buyer segment. After SR-GNN intervention, the Browser segment sharply declined to 18.4%. Most of this volume shifted toward the Bouncer segment (35.2% $\rightarrow$ 77.7%), which is a direct consequence of the simulation mechanism: sessions successfully guided toward relevant products earlier by SR-GNN naturally terminate before crossing the 5-view threshold and therefore fall into the Bouncer category by definition. Quick Buyer increased slightly from 1.3% to 2.3%, reflecting additional short sessions with successful purchases enabled by accurate early-stage recommendations.

The central positive signal is that SR-GNN transforms a substantial proportion of browse-without-purchase sessions into early-ending sessions. Instead of continuing to browse through 5+ products without purchasing, users either find suitable products earlier (shorter sessions) or convert directly within short sessions (increase in Quick Buyer). This provides direct evidence that SR-GNN reduces friction at deeper interaction stages.

C. AOV Proxy Analysis

Figure 7 reports an increase in AOV proxy from 865.0 (Before) to 976.1 (After), corresponding to a +12.8% improvement. This result is consistent with the observation that SR-GNN’s Revenue-weighted HR@20 exceeds standard HR@20, confirming that the model’s accuracy advantage is not uni-

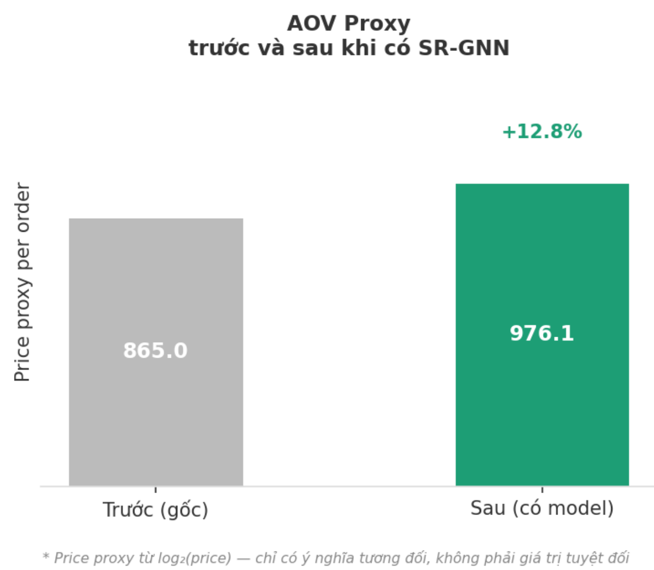

Fig. 7: AOV proxy before and after SR-GNN intervention.

formly distributed across price levels but is more concentrated on higher-value products.

VI. Conclusion

Many online shoppers browse without logging in, which makes it hard for traditional recommendation systems to suggest relevant products. This study shows that we can still make good recommendations using only what a user does within a single session — no account or history needed. We tested three models on the Diginetica e-commerce dataset. The simplest model, which just recommends popular items, performed poorly. A co-occurrence model did better by looking at what item a user viewed last. But SR-GNN — a graph neural network that reads the entire sequence of viewed items — performed best by a wide margin, correctly predicting the next item in about 1 out of every 2 sessions (HR@20 = 0.54). Beyond accuracy, simulations suggest that better recommendations lead to real business benefits: higher conversion rates, fewer sessions ending without a purchase, and a roughly 13In short, even when we don’t know who the user is, we can still understand what they want — and serve them better for it.